Know Your Story

Module Two

Instructions

In this course, we will use the attached article as a case study for creating a presentation from a research project. At a minimum, you should read the abstract, introduction and conclusion to familiarize yourself with the research topic and key terms.

Use the following reading comprehension questions to check that you have a good grasp on the material:

1. What two types of measures do the authors mention for observing liquidity in a market?

2. Why are some investors limited to using proxies for observing liquidity?

3. What market did the authors focus on in their analysis?

4. Which proxy did the authors find performed best in low-liquidity environments?

See the last page in this document to check your answers.

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What is the best proxy for liquidity in the presence of extreme illiquidity?

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ABSTRACT

This paper aims to indicate the best daily proxy for unobserved liquidity in the presence of extreme movements on the market. We apply copulas to investigate the dependence between benchmarks based on intraday prices and proxies based on daily data. We focus on the tail dependence between both types of measures. Our results show that when the market experiences extreme illiquidity, the Closing Quoted Spread (CQS) based on daily closing bid and ask prices is superior to other percent-cost low-frequency proxies. We find the highest tail dependence coefficients for CQS and either the Percent Effective Spread or the Percent Quoted Spread.

1. Introduction

As stated in the seminal paper of Kyle (1985) liquidity is an elusive concept. Thus, its estimation and approximation are considered to be very challenging tasks. There are many concepts and approaches presented in the available literature, but still none of them has been indicated to be the most suitable one (Goyenko et al., 2009; Fong et al., 2017). Without access to transaction data there may be no agreement on the issue which measure is better, to what extent these measures are coherent and how well they reflect unobservable liquidity. Generally, two classes of measures are considered: benchmarks that are calculated on the basis of high-frequency data, and proxies that are based on low-frequency, typically daily data. Even today high-frequency data are not collected for some markets. If they are, databases are usually expensive, which restricts their accessibility for investors. Thus it is crucial to find accurate liquidity proxies based on widely available data, which could be used instead of the high-frequency benchmarks. A range of measures for both classes have already been proposed (Díaz and Escribano, 2020) and their ability to reflect elusive liquidity have been examined (Corwin and Schultz, 2012; Goyenko et al., 2009; Fong et al., 2017). The literature on the coherence of liquidity proxies with benchmarks already seems to be exhaustive, but it is limited to the developed markets (Fong et al., 2017; Goyenko et al., 2009; Corwin and Schultz, 2012), with only few studies devoted to the emerging markets (Batten and Vo, 2014; Kang and Zhang, 2014; Olbryś, 2018; Będowska-Sójka, 2018).

Studies carried out so far, either on the developed or the emerging markets, focus on the coherence of the measures estimated as correlation coefficients and usually neglect dynamics of these relations (Lesmond, 2005; Goyenko et al., 2009; Corwin and Schultz, 2012; Díaz and Escribano, 2020). However, the proxies that in general perform well may behave differently in the presence of extreme liquidity movements in the market. This issue is important for risk management or portfolio management. Although extreme movements are by definition rare, they are of utmost importance for investors (Neftci, 2000). The rationale for analysis of extreme market conditions stems from the fact that investors are not concerned with liquidity itself when the financial market is stable. In contrast, these issues rapidly gain in importance when the market becomes more volatile and the problem of illiquidity arises. A number of studies shows that in turbulent times financial institutions withdraw their funds and move them to the so-called "safe

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havens" (Rösch and Kaserer, 2014; Giot et al., 2010; Easley et al., 1996; Nagel, 2012; Chung and Chuwonganant, 2018).

The aim of our paper is to examine how adequate percent-cost liquidity proxies prove to be in the presence of extreme market illiquidity. This study follows the methodology used in a survey of Fong et al. (2017), as we compare benchmarks calculated on the basis of high-frequency data with proxies based on daily data. It is conducted on the Warsaw Stock Exchange (henceforth WSE), which has experienced rapid growth within the last decade. In the paper a unique tick-by-tick database created from data obtained directly from the WSE is used. Analyses are conducted on all 141 stocks that have been continuously listed from 2006 to 2016. Within this 11-year sample period financial markets experienced both stable and turbulent times. This unique dataset facilitates examination which proxies show the highest dependence with benchmarks when liquidity evaporates. It is also analysed whether firm size influences the strength of the dependence between proxies and benchmarks.

Our contribution to the literature on the subject is the following: firstly, we find that the liquidity measure proposed by Chung and Zhang (2014), the Closing Percent Quoted Spread (CQS), is the best among the considered percent-cost proxies. This property holds when liquidity is measured in extreme situations (e.g. when liquidity is abnormally low). In order to demonstrate this, we use copulas and estimate tail dependence coefficients. Thus instead of estimating linear dependencies, which is a common approach adopted in the literature (Kang and Zhang, 2014; Lesmond, 2005; Goyenko et al., 2009), we focus on extreme values. That might be especially important on the emerging markets, where changes in returns and liquidity are substantial (Bekaert and Harvey, 1997). To our best knowledge this is the first such study which measures the tail dependency between liquidity proxies and benchmarks.

Secondly, these findings enhance our understanding of liquidity measurement by considering stocks listed on the WSE, which was an emerging market at that time (since September 2018 FTSE Russell has promoted Poland to the Developed Market status). We show that the linear dependency between proxies and benchmarks obtained for developed markets is similar to the one observed on the emerging market. Thirdly, we find that if bid and ask quotes at the end of the day are not available (and CQS may not be calculated), the High-Low Range may be used instead of the Effective Spread Estimator of Corwin and Schultz (2012) as a daily liquidity proxy. The tail dependence coefficients for both measures are similar, but the non-parametric test for the existence of extreme dependency of Falk and Michel (2006) speaks for the High-Low Range. Fourthly, we show that although spreads are higher for small stocks than for big ones, the size of the company has no impact on the nature of dependency between a benchmark and a proxy. Also, we find that the introduction of the Universal Trading Platform in 2013 has led to deterioration of liquidity in the case of small companies. These results carry implications for both researchers and practitioners. Without going through the time-consuming handling of high-frequency datasets, we may use a simple proxy, proven to be suitable for the developed markets (Chung and Zhang, 2014). It is also the best proxy for relatively less examined emerging markets, even in crisis times.

The rest of the paper is the following: in Section 2 the methodology is described, which includes liquidity benchmarks and proxies used in the study as well as the description of copulas employed in the empirical part. Section 3 presents data used in the study, their transformation and aggregation. Section 4 provides the results of empirical research, while Section 5 gives concluding remarks and indicates some further issues.

2. Methodology

2.1. Liquidity benchmarks and proxies

In the paper we focus on finding the best liquidity proxy based on the daily data from the Polish capital market. We consider mainly relations of proxies and benchmarks in the case of extreme market movements, when liquidity evaporates rapidly. We also investigate, if the results obtained for the whole market are robust with respect both to the size of a company and different subperiods.

There are two main categories of daily liquidity measures. The first is related to trading volume and thus is called cost-per-dollar-volume (Fong et al., 2017). This approach has some shortcomings, as the figures are past-oriented and numbers reflect activity rather than liquidity itself (Lesmond, 2005). Also, volume data are said to be noisy from the statistical point of view (Lee and Swaminathan, 2000). In another, parallel approach, liquidity is perceived as a direct benefit for market participants, which comes together with low transaction costs. For a highly liquid security these costs should be lower than for an illiquid asset. The percent-cost measures are related to the bid-ask spread, which is the difference between the best buy and the best sell offer within a certain span of time. Spreads are directly tied to liquidity – the lower the spread is, the more liquidity is offered. Investors benefit from liquid assets, as these facilitate minimisation of losses resulting from any opening and closing of a position (Chordia et al., 2000).

In our paper we focus on percent-cost measures and thus our search concerns only that specific dimension of liquidity. Four benchmarks used in Fong et al. (2017) are considered - they are focused on some combination of transaction prices or order quotes, but cover slightly different notions of liquidity:

- Percent Effective Spread:

$$PES_k = 2D_k(\log(P_k) - \log(M_k)), \tag{1}$$

where P_k is the price of the last transaction in a 5-minute time interval k, while M_k is the mid price of the best ask quote, P_A, and the best bid quote, P_B, within a given interval; $M_k = 0.5(P_{A_k} + P_{B_k})$. D_k is an indicator variable showing the direction of the *k-th* trade with 1 and -1 for buy and sell orders, respectively. In order to indicate the direction of a trade, we apply the Lee and Ready algorithm (Lee and Ready, 1991). The *PES* are then aggregated into daily spreads as weighted averages, where weights depend on the volume of orders. The aggregation is done for each stock separately on daily and monthly basis by calculating volume-weighted average. The

same procedure is applied to the other benchmarks.

- Percent Price Impact:

$$PPI_k = 2D_k(\log(M_{k+5}) - \log(M_k)). \tag{2}$$

where M_{k+5} stands for the mid price of a transaction in the next time interval (five minutes). The *PPI* shows the impact of quotes on spreads.

- Percent Quoted Spread:

$$PQS_k = 2(P_{A_k} - P_{B_k})/(P_{A_k} + P_{B_k}). \tag{3}$$

This spread is also based on quotes only. *PQS* is commonly used in the literature, sometimes under different names, e.g. in Boudt and Petitjean (2014) it is called the proportional quoted bid-ask spread.

- Percent Realised Spread:

$$PRS_k = 2D_k(\log(P_k) - \log(M_{k+5})). \tag{4}$$

This spread measure estimates upcoming transaction costs.

Out of the large collection of liquidity proxies in the literature, we select these measures that prove to be most successful on the developed markets (Corwin and Schultz, 2012; Chung and Zhang, 2014; Karolyi et al., 2012; Fong et al., 2017), are non-parametric and to calculate which only daily four prices (high, low, open and close) are required. In order to relate them to the spread benchmarks, the chosen proxies are percent-cost only:

- Effective Spread Estimator of Corwin and Schultz (2012)

$$CSA_t = \frac{2(e^{\alpha_t} - 1)}{1 + e^{\alpha_t}}, \tag{5}$$

where $\alpha_t = \frac{\sqrt{2\beta_t} - \sqrt{\beta_t}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma_t}{3 - 2\sqrt{2}}}$, $\beta_t = \left(\log\left(\frac{H_t}{L_t}\right)\right)^2 + \left(\log\left(\frac{H_{t+1}}{L_{t+1}}\right)\right)^2$, and $\gamma_t = \log\frac{max(H_t, H_{t+1})}{min(L_t, L_{t+1})}$. Following Corwin and Schultz (2012) we adjusted the spread estimator for overnight returns.

- Closing Percent Quoted Spread

$$CQS_t = 2(P_{A_t} - P_{B_t})/(P_{A_t} + P_{B_t}), \tag{6}$$

where P_{A_t} and P_{B_t} are ask and bid quotes, respectively, observed at the end of day t.

High-Low Range

$$HLR_t = 2(H_t - L_t)/(H_t + L_t), \tag{7}$$

where H_t and L_t are the high and the low prices within the day.

These benchmarks account for different aspects of liquidity: the Effective Spread Estimator *CSA* and the Closing Percent Quoted Estimator *CQS* are measures of transaction costs. The last proxy, the High-Low Range, was used previously in Będowska-Sójka (2019) as an adjusted form of the *CQS* measure. The bid and ask quotes at the end of the day are not provided in the widely available datasets, but as shown by Corwin and Schultz (2012), the high price within a day is a buyer-initiated (ask) trade, whereas the low price is rather the seller-initiated (bid) trade. Thus replacing ask and bid prices with the high and low ones appears to be a feasible extension. With our data we are able to provide evidence how well this approximation works with respect to the original *CQS* measure.

2.2. Copulas and tail dependence coefficient

In the theory of probability and statistics copula C is a multivariate probability distribution, which is used to describe the dependence between random variables apart from their marginal distribution. An extensive introduction into copulas can be found in Nelsen (2006) and Cherubini et al. (2006). This methodology has recently become a significant new tool to handle the co-movement between markets and as such it is used in risk management, portfolio optimisation, and other relevant issues considered in finance (Cherubini et al., 2006). In our study a two-dimensional copula is considered. Such a copula joins marginal distributions of random variables X_1 and X_2 to form a bivariate distribution function as follows:

$$F(x_1, x_2) = C(F_1(x_1), F_2(x_2)), \tag{8}$$

where $F_j(x_j)$ for $j = 1,2$, are marginal distribution functions of X_1 and X_2 respectively, while $F(x_1, x_2)$ is a two-dimensional distribution function of X_1 and X_2. Sklar's Theorem (Sklar, 1959; Vrac et al., 2012) guarantees the existence and uniqueness of a copula for continuous random variables.

The copula is a highly flexible tool allowing to handle changes of variables occurring together. It is useful especially when the dependence between extreme movements, such as extreme illiquidity, is considered. The concept of tail dependence coefficients

(TDC), readily obtainable from a copula, is the basic measure of a dependence between extreme values of random variables. Generally speaking, a tail dependence describes the limiting proportion, by which one margin exceeds a certain threshold, provided that the other margin has already exceeded that threshold. Because extreme values may appear both in the left and the right tail of distribution, there are two types of TDC – lower and upper ones. We focused on extreme illiquidity in this study, therefore only the upper TDC is analysed here.

Formally, the upper tail dependence coefficients is defined as (Coles et al., 1999):

$$\lambda_U = \lim_{v\to 1} P(X_1 > F_1^{-1}(v) \mid X_2 > F_2^{-1}(v)) = \lim_{v\to 1}(1 + 2v + C(v,v))/(1 - v), \tag{9}$$

provided that the above limit exists. If TDC is equal to zero, $\lambda_U = 0$, variables X_1, X_2 are said to be asymptotically independent in the upper tail of distribution. In the case of $\lambda_U \in (0,1]$, there exists an extreme dependence: the closer the value of TDC is to one, the stronger the dependence.

There are many parametric copulas proposed in literature. The most popular in finance applications are Archimedean, elliptical and extreme-value copulas (Cherubini et al., 2006; Nelsen, 2006). In this paper we applied the Gumbel, Joe and Hüsler-Reiss copulas. This selection stems from the fact that all of them indicate only the right dependence property, which is desired in our study.

The Gumbel copula belongs to both the Archimedean and extreme-value copula families. It is of the form:

$$C_\theta^{Gu}(u,v) = \exp(-((-\log u)^\theta + (-\log v)^\theta)^{1/\theta}),\ \theta \in [1,\infty). \tag{10}$$

For $\theta = 1$ random variables are independent, whereas for $\theta \to \infty$ they are co-monotonic (i.e. show a positive monotone dependence). The Gumbel copula has the upper tail dependence equal to $\lambda_U = 2 - 2^{1/\theta}$.

The Joe copula belongs to the Archimedean copula family. It is of the form:

$$C_\theta^{Jo}(u,v) = 1 - ((1-u)^\theta + (1-v)^\theta - (1-u)^\theta(1-v)^\theta)^{1/\theta},\ \theta \in [1,\infty). \tag{11}$$

For $\theta = 1$ the independence is obtained, whereas for $\theta \to \infty$ the complete asymptotic dependence is obtained. The Joe copula has the upper tail dependence equal to $\lambda_U = 2 - 2^{1/\theta}$.

In turn, the Hüsler-Reiss copula belongs to the extreme-value copula family. It is of the form:

$$C_\theta^{HR}(u,v) = \exp(-\hat{u}\Phi(1/\theta + \theta/2\log(\hat{u}/\hat{v})) - \hat{v}\Phi(1/\theta + \theta/2\log(\hat{v}/\hat{u})),\ \theta \in (0,\infty), \tag{12}$$

where: $\hat{u} = -\log(u)$, $\hat{v} = -\log(v)$, and Φ is the standard normal CDF. For $\theta \to 0$ random variables are independent, whereas for $\theta \to \infty$ they are co-monotonic. The Hüsler-Reiss copula has the upper tail dependence equal to $\lambda_U = 2 - 2\Phi(1/\theta)$.

In order to estimate parameter θ, the maximum likelihood method is used. Hereby we obtain a consistent and asymptotically normal estimation of θ. When testing for the existence of an extreme dependence between two variables, it is typically examined whether there exists any structure of extreme dependence. The popular procedure applied in such situations comes from Falk and Michel (2006). Asymptotic independence is tested by the authors using four different statistics i.e. Neyman-Pearson, Fisher, Kolmogorov-Smirnov and Chi-square. Numerous simulations indicate that the Neyman–Pearson test has the smallest type II error rate (Falk and Michel, 2006).

Suppose we have n independent copies of $(X_{1,\ 1}, X_{2,\ 1})$, …, $(X_{1,\ n}, X_{2,\ n})$ of random vector (X_1, X_2). Fix $c < 0$ and consider only those observations $X_{1,\ i}$, $X_{2,\ i}$ among the sample that satisfy $X_{1,\ i} + X_{2,\ i} > c$. Denote these by C_1, …, C_m in the order of their outcome. One has to decide whether the distribution function of $W_i = C_i/c$ is equal to either the null hypothesis $F_0(t) = t^2$ or the alternative $F_1(t) = t$, $t \in [0,1]$. The Neyman-Pearson test statistics is as follows:

$$NP = -\sum_{i=1}^{m}(\log(W_i) - m\log(2)). \tag{13}$$

And the p-value of the test is:

$$p_{NP} = \phi\left(\frac{2\sum_{i=1}^{m}\log(W_i) + m}{\sqrt{m}}\right). \tag{14}$$

3. Data

We use the tick-by-tick data on 141 stocks listed on the Warsaw Stock Exchange continuously from January 2006 to December 2016. The Warsaw Stock Exchange (WSE) is an order-driven market with continuous trading. There exists a continuous double auction mechanism; submitted orders are displayed in the order book and matched automatically. Thus the market operates without any market makers, and orders are matched based on the price and time priority. The trading hours changed several times within our period of analysis and we accounted for these changes when calculating spread benchmarks. In December 2016 the exchange was working from 9:00 AM to 4:50 PM (CET).

Our sample consists of 42 big firms, 37 medium and 62 small. These companies are categorised to the size groups according to the rules applied on the WSE in the following manner: the 'big' label is reserved for these firms that have capitalisation over than 250 million Euro, 'medium' are these with capitalisation of 50 to 250 million, while 'small' stocks are those that have capitalisation below 50 million Euro. In order to simplify the categorisation into different size groups the capitalisation of stocks at the end of 2016 is

applied.

Cross-sectional studies, where proxies from many countries are considered, typically suffer from limitations due to a number of stocks from each market and may result in losing insightful information on companies that differ by size and trading volume. Division of the stocks into different size groups facilitates determination: (1) if spreads for stocks of various sizes are similar, and (2) if there are any differences in the dependence between benchmarks and proxies for stocks that differ by size. With respect to the first issue, it is expected that benchmarks (and proxies) for big stocks will be lower than those for small stocks. This is based on the empirical fact evidenced in the literature, that big stocks are usually more liquid and characterised by lower liquidity premiums (Mancini et al., 2013). With respect to the second issue, hypothesis is proposed that big stock proxies should demonstrate a greater dependency on benchmarks than small stock proxies. This might be driven by the tendency of financial intermediaries to invest in blue chip stocks (and is illustrated by the portfolio of indices such as e.g. MSCI or Stoxx). Our database is built on the data from the electronic limit order book operating in the Universal Trading Platform (UTP) system. We use tick-by-tick data for the whole sample period, comprising prices and volumes of each trade as well as the best bid and ask offers in the order book during a trading day. Our database includes both trades and best orders in the limit order book. Analysis are conducted on orders made within the session, as well as those placed at the time of an auction (fixing) at the beginning and the end of the day. The raw data are filtered according to the procedure described in Barndorff-Nielsen et al. (2009).

We construct the database for equally sampled 5-minute data and on this basis calculate liquidity benchmarks, which are then aggregated into daily and monthly values. This is done for each stock separately both over the whole sample period and subperiods.

4. Empirical results

The empirical part is divided into two sections: the first one focuses on the descriptive statistics of benchmarks and proxies over the whole sample period. The main hypothesis in this part states that liquidity measures are time-varying and their values depend on the size of the company. The second part is devoted to the examination of tail dependence coefficients, which allows us to describe relations between measures of liquidity in the case of extreme illiquidity. We use copulas to identify characteristics of these relations.

4.1. Descriptive statistics of liquidity measures over the whole period

Table 1 shows descriptive statistics for benchmarks and proxies in daily frequency. If one considers spreads for all stocks, the mean values of particular measures differ, ranging from 0.67 (PPI) to 3.41 (HLR). The differences in magnitude of average liquidity proxies within a sample are natural as these measures gauge different concepts of calculating transaction costs. Some features are common: for all measures the means are higher than medians, which suggests that the distributions of the measures are skewed to the right and some extreme values of the measures appear within the period. The asymmetric feature of the distribution is observed also for different size groups. Among all the variables HLR seems to be an exception, as this is almost two-fold greater than the second highest value observed for CQS. The spreads are the lowest for the big stocks, and the highest for the small stocks. This confirm our hypothesis, that bigger stocks are more liquid and thus should have lower transaction costs (lower spreads) than smaller stocks. It also shows that differentiating stocks by size is reasonable.

Then the liquidity measures are investigated as aggregated into monthly benchmarks and proxies. The descriptive statistics for these values are presented in Table 2. The distribution of the liquidity measures seems to be asymmetric - in most cases the median is higher than the mean, excluding PQS. Similarly as for daily measures, the highest mean value is observed for HLR, while the lowest is recorded for PPI. When one considers different size groups the highest spreads are observed for the small stocks, while the lowest for the big stocks - the latter are 2–3 times smaller than the former. When daily and monthly measures are compared, the differences between the values are in most cases small - the highest difference is observed for CSA with the daily mean for all stocks of 0.91 versus the monthly mean for all stocks of 1.52.

Next, we show the dynamics of both classes of measures, benchmarks and proxies, within the whole sample period. Fig. 1 presents the dynamic behavior of the monthly benchmarks (top panel) and proxies (bottom panel) within the time. The values are calculated on the basis of averages observed in each month for the whole sample of stocks.

The top panel shows four benchmarks obtained for the sample of stocks in monthly frequency: PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). In the bottom panel HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

Although there are some differences in the particular values, the overall dynamic of the spreads is similar. In 2006 when sample starts spreads are relatively low, then they almost double as a result of the global financial crisis, GFC, in 2008 and decrease in the mid 2009. Next, an increase is observed in mid 2011 and is related to the European sovereign debt crisis. Surprisingly, the highest values across the whole sample are recorded in April 2013 - there are no specific events recognized either globally or in the European region that could explain that sharp move in spreads (Będowska-Sójka and Echaust, 2019). However, on 15 April 2013 the WSE introduced a new trading system, the Universal Trading Platform, UTP. All spreads, except for PPI, increase to significantly higher levels and remain so until early 2014. Then they decreased back to the 2011 level showing no pronounced movement until the end of the sample.

The bottom panel of Fig. 1 presents the behaviour of liquidity proxies, that is consistent with the one observed for benchmarks. The proxies increase in value in the time of GFC, then decrease until the European sovereign debt crisis, which is distinguished by a single peak. Then they increase again in 2013, but to a varying extent: HLR and CSA increase relatively less than the benchmarks,

Table 1
Descriptive statistics of daily liquidity proxies and benchmarks for all companies and different size groups.

	Minimum	1Quantile	Median	Mean	3Quantile	Maximum
			All firms			
PES	0.63	1.14	1.41	1.47	1.71	3.13
PPI	0.20	0.47	0.61	0.67	0.80	2.93
PQS	0.73	1.15	1.50	1.54	1.79	3.75
PRS	0.34	0.78	1.07	1.12	1.34	3.04
HLR	1.53	2.67	3.19	3.41	3.83	13.66
CSA	0.24	0.70	0.86	0.91	1.03	3.96
CQS	0.57	1.18	1.52	1.58	1.82	4.15
			Big firms			
PES	0.33	0.61	0.72	0.75	0.86	2.43
PPI	0.08	0.26	0.35	0.39	0.48	1.45
PQS	0.41	0.66	0.78	0.81	0.92	2.97
PRS	0.18	0.41	0.51	0.54	0.63	2.45
HLR	1.20	2.14	2.51	2.75	3.05	10.02
CSA	0.09	0.50	0.62	0.67	0.77	3.51
CQS	0.31	0.64	0.75	0.80	0.89	5.61
			Medium firms			
PES	0.61	1.10	1.31	1.38	1.62	4.65
PPI	0.12	0.41	0.54	0.60	0.72	3.32
PQS	0.69	1.13	1.37	1.44	1.67	5.89
PRS	0.34	0.74	0.97	1.05	1.26	3.36
HLR	1.39	2.58	3.11	3.34	3.81	12.34
CSA	0.04	0.63	0.82	0.87	1.03	4.46
CQS	0.38	1.11	1.35	1.43	1.67	7.16
			Small firms			
PES	0.65	1.46	1.93	2.00	2.35	5.10
PPI	0.18	0.59	0.79	0.90	1.07	5.36
PQS	0.78	1.45	2.02	2.09	2.45	6.04
PRS	0.33	1.00	1.44	1.56	1.91	5.25
HLR	1.45	3.00	3.63	3.89	4.49	17.39
CSA	0.27	0.78	1.00	1.10	1.27	4.36
CQS	0.65	1.51	2.08	2.19	2.58	7.33

Note: Descriptive statistics (multiplied by 10^2) are presented for liquidity benchmarks and proxies calculated as the aggregate equally-weighted portfolio of stocks. Liquidity proxies are averaged within a given day and there are 2736 days. PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

while CQS experiences an increase comparable to that in 2008.

The increase in the benchmarks in 2013 seems to be counter-intuitive: the new system introduced to the WSE aimed at enhancing liquidity on the market and thus should decrease spreads. A partial explanation to this surprising result lies in the size of the companies. In Fig. 2 the behaviour of four benchmarks is shown for three size groups separately. The differences in the benchmarks' behavior for the big and the small firms are remarkable. The particular increase in spreads in 2013 is due to the small and medium firms rather than to the big ones and could be motivated by the lack of interest of high-frequency investors in the small and medium companies on the WSE. The high volatility providers and high liquidity fund programmes were implemented as late as in 2016 and only then could potentially attract high-frequency traders.[1] Another explanation can be found in co-movement between liquidity and volatility which is documented in many studies (Chung and Zhang, 2014; Ma et al., 2016; Będowska-Sójka and Echaust, 2019). As HLR and CSA proxies do not indicate strong dependencies with PES, PQS or PRS benchmarks (see table A1), the former may be much more related to volatility than to liquidity. Thus an increase in illiquidity in 2013 is reflected in benchmarks (especially those calculated for small and medium firms), but not in proxies, that are based on daily prices.

This figure shows four benchmarks obtained for the sample of stocks in monthly frequency: PES stands for Percent Effective Spread, PPI is Percent Price Impact, PQS denotes Percent Quoted Spread, while PRS is Percent Realised Spread (Fong et al., 2017). Firm size are considered in three categories: big, medium and small.

A figure showing daily liquidity measures is given in the Appendix (Fig. A1). Both liquidity proxies and spreads exhibit some discontinuities and sharp jumps (in daily as well as monthly data). In that aspect, liquidity measures are similar to volatility estimates, which are often characterised by comparatively large movements of a given variable in a short period of time (the so-called spikes). These jumps are usually described as liquidity shocks (Lahaye et al., 2011).

[1] https://www.gpw.pl/incentive_schemes.

Table 2
Descriptive statistics of monthly liquidity proxies and benchmarks for all companies and different size groups.

	Minimum	1Quantile	Median	Mean	3Quantile	Maximum
			All firms			
PES	0.75	1.18	1.45	1.49	1.72	2.61
PPI	0.31	0.53	0.63	0.68	0.81	1.47
PQS	0.84	1.21	1.61	1.60	1.86	2.93
PRS	0.47	0.66	0.96	0.98	1.20	2.13
HLR	2.62	3.07	3.66	3.81	4.27	7.46
CSA	0.99	1.25	1.48	1.52	1.68	2.60
CQS	0.91	1.30	1.77	1.79	2.12	3.74
			Big firms			
PES	0.43	0.64	0.73	0.74	0.83	1.26
PPI	0.14	0.24	0.32	0.36	0.44	0.77
PQS	0.49	0.69	0.82	0.81	0.93	1.29
PRS	0.22	0.36	0.45	0.47	0.54	1.04
HLR	2.01	2.33	2.63	2.84	3.10	6.51
CSA	0.74	0.91	1.01	1.11	1.23	2.36
CQS	0.54	0.70	0.80	0.83	0.90	1.47
			Medium firms			
PES	0.78	1.06	1.25	1.34	1.57	2.41
PPIt	0.26	0.45	0.55	0.61	0.72	1.89
PQS	0.90	1.13	1.33	1.42	1.65	2.80
PRS	0.43	0.60	0.80	0.84	0.97	1.68
HLR	2.29	2.96	3.42	3.61	4.07	6.55
CSA	0.92	1.19	1.40	1.44	1.60	2.63
CQS	0.90	1.17	1.45	1.53	1.78	2.71
			Small firms			
PES	0.80	1.52	2.11	2.08	2.49	4.08
PPI	0.42	0.71	0.86	0.95	1.14	2.27
PQS	0.96	1.49	2.23	2.23	2.60	4.46
PRS	0.47	0.85	1.32	1.41	1.78	3.92
HLR	3.08	3.66	4.45	4.60	5.20	8.74
CSA	1.14	1.50	1.75	1.84	2.10	3.07
CQS	1.06	1.67	2.61	2.61	3.05	6.34

Note: Descriptive statistics (multiplied by 10^2) are presented for liquidity benchmarks and proxies calculated as the aggregate volume-weighted average measures within a market. A number of months across which the average is calculated is 131. PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

4.2. In search of the best proxy: tail dependence coefficient

The main purpose of our paper is to examine which percent-cost liquidity proxy is the best one when liquidity evaporates. We employ the copula methodology to study interdependencies between benchmarks and proxies. Similarly to Fong et al. (2017) we use two parallel approaches. The first one is based on the Fama and MacBeth (1973) procedure. For each day or month the cross-sectional tail dependence coefficient is calculated across all stocks and then the average coefficients over all days and months are obtained.

The second approach is the dynamic one and is based on equally-weighted portfolios across all stocks for day k. That is, a portfolio of proxies and benchmarks is computed in day k by taking the average of detrended liquidity measures over all stocks in day k. This approach is called the time-series for portfolios. All benchmarks and proxies are calculated for each stock, the differences of the series are determined and the equally-weighted portfolios are built for each day. Then the dependency between different portfolios, each for a single measure, is estimated. We supplement this approach by applying an asymptotic nonparametric independence test as a robustness check for the existence of a dependence between any pair of a benchmark and a proxy. All three proxies and four benchmarks are calculated also in monthly frequency - they are averaged across a given month. Then the same procedure is repeated to obtain tail dependence coefficients for monthly liquidity measures. Finally the tail dependence coefficients are analysed in three subperiods and thus robustness of our results is verified.

4.2.1. Cross-section dependency

Table 3 presents average values of tail dependence coefficient, λ, in a cross-section. Here we present the results for the Gumbel copula only. In the case of daily data the highest tail dependence coefficients are observed for CQS. Thus that proxy is indicated as the best one among all considered percent-cost benchmarks and this result holds for all size groups. The tail coefficients with benchmarks differ substantially: the highest values of coefficients are observed independently of the size group for PQS, followed by PES. The difference between the coefficients in the case of these two benchmarks is small and does not exceed 2 p.p. (λ's are around 60%). Next is PRS with values of the λ coefficients close to 50%, and the last among percent-cost benchmarks is PPI with much lower coefficients

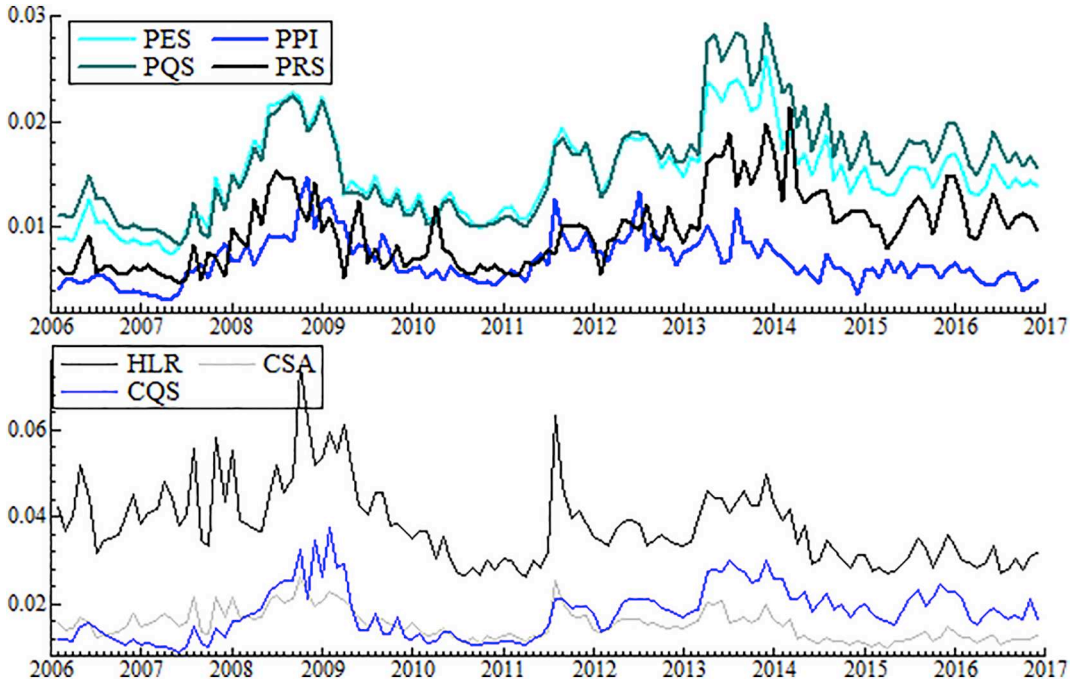

Fig. 1. Monthly spread measures.

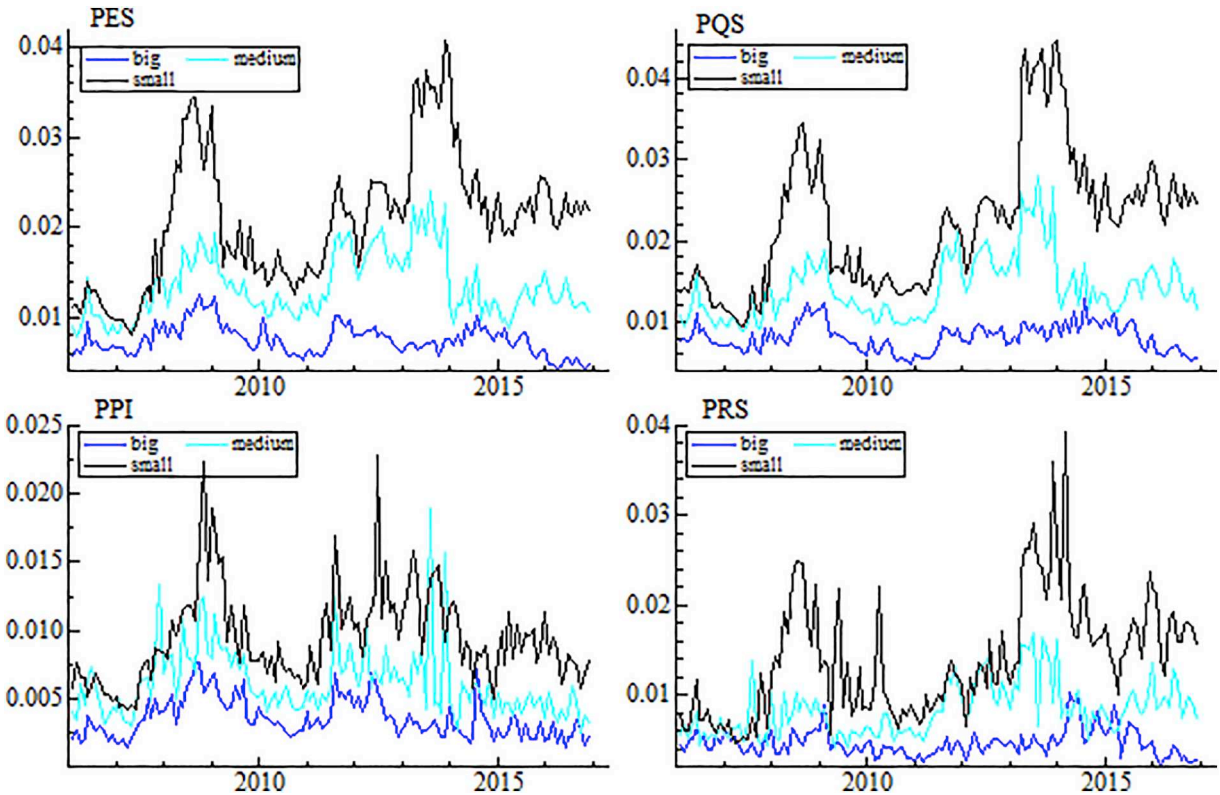

Fig. 2. The benchmarks differentiated by firm size.

Table 3
Tail dependence coefficients: Average cross-section.

	Daily			Monthly		
	HLR	CSA	CQS	HLR	CSA	CQS
			All firms			
PES	22.66	21.43	62.50	26.09	24.36	77.82
PPI	24.20	12.61	35.00	26.80	21.51	50.19
PQS	25.33	23.35	65.35	29.27	27.72	79.79
PRS	20.63	21.00	52.82	21.54	20.66	62.86
			Big firms			
PES	10.45	10.12	58.25	6.34	4.37	77.32
PPI	18.69	12.33	38.16	11.61	7.36	55.37
PQS	12.42	11.64	60.44	7.72	6.12	79.08
PRS	10.63	12.19	47.88	6.51	5.40	63.77
			Medium firms			
PES	21.38	22.96	56.91	16.94	18.36	71.61
PPI	24.44	17.30	28.28	19.18	17.02	38.64
PQS	23.66	24.48	59.68	18.14	20.18	73.91
PRS	19.51	22.86	48.55	13.38	15.77	57.17
			Small firms			
PES	24.43	24.75	61.08	27.07	25.68	74.82
PPI	24.77	14.85	29.05	24.96	19.18	41.39
PQS	27.66	27.05	64.89	31.05	30.11	77.32
PRS	22.46	24.44	51.85	23.04	23.11	60.73

Note: The tail dependence coefficients between proxies and benchmarks (expressed as a percantage) are estimated for daily and monthly data within the Gumbel copula. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

(below 40% or even 30%).

If bid and ask prices at the end of the day are not available and CQS cannot be calculated, there is no clear answer, which of the two remaining proxies is better. For all stocks HLR has a higher value of the λ coefficient in three out of four cases, but this advantage is not stable for stocks divided into size groups. In the case of PPI it is always better to take into account HLR rather than CSA.

In the case of monthly data the λ coefficients behave in the same manner, but they are higher than for daily ones. For CQS λ is as high as 0.798 (with PQS) or 0.778 (with PES). Again, CQS-PES and CQS-PQS are the best pairs, with the highest tail dependence coefficients. When stocks are divided into size groups, CQS as a proxy displays a significantly stronger dependency than CSA or HLR in each size group. Moreover, for the latter proxies, the coefficients for small stocks are higher than for big stocks. At this stage it is difficult to indicate unambiguously which of these two is a better proxy.[2]

4.2.2. Portfolio time-series approach

Table 4 presents values of dependence coefficients calculated for the time series of proxies and benchmarks. The presented results are provided by the Gumbel copulas. Some characteristics of the λ coefficient in this approach are similar to these observed in the cross-section. CQS is the best liquidity proxy at the time of extreme illiquidity. This result is stable in daily and monthly data, although the dependency coefficients for monthly data are generally higher than for daily data. They are also stable across different size groups. The highest values of the λ coefficients are obtained for the same proxy-benchmark pairs, CQS-PES and CQS-PQS, but they are lower than for cross-section. As the time-series approach is applied for differences in liquidity measures, the results might differ from those observed for the average cross-section. Few differences are mostly remarkable: the CSA proxy in daily data shows very low coefficients for all benchmarks. This result changes significantly when monthly data are considered - here CSA is often better than HLR. Further, the daily HLR proxy has a very low tail-dependence coefficient with PRS for all size groups. Also, HLR among all the three percent-cost proxies has the highest tail-dependence with the PPI benchmark. Last but not least, in the monthly data the distance between the best proxy, CQS, and the other two, HLR and CSA, diminished. In the time series approach both proxies, HLR and CSA, have higher λ coefficients with PES and PQS than in the cross-section, and thus they are closer to the winner of the race, CQS.[3]

[2] In the Appendix we showed Spearman correlation coefficients for proxies and benchmarks, see table A1. The general result is the same: CQS is the best proxy, while the best pair for CQS is that with PES or PQS, both in daily and monthly data. The low tail dependence corresponds to a negative or zero correlation, while the high tail dependence is observed in the cases where we have high correlation coefficients. Note that the correlation coefficient falls within the range of $[-1;+1]$ and focuses on linear dependence, while the tail dependence coefficient is from the interval $\lambda_U \in [0,1]$ and captures the nonlinear dependence of extreme values only.

[3] For time series we show the Spearman correlation coefficients for proxies and benchmarks in the Appendix, see table A2. The general result is not changed: CQS is the best proxy, while the best pair for CQS is that with PES or PQS.

Table 4
Tail dependence coefficients: Portfolio time-series approach.

	Daily			Monthly		
	HLR	CSA	CQS	HLR	CSA	CQS
			All firms			
PES	16.83	0.22	43.52	39.81	44.23	55.62
PPI	16.11	0.00	6.25	21.91	21.47	16.26
PQS	18.12	0.21	46.61	43.01	46.60	56.92
PRS	8.59	1.46	19.84	21.03	27.54	35.68
			Big firms			
PES	13.52	2.92	38.98	45.18	43.04	49.69
PPI	11.15	0.00	9.41	13.85	11.75	18.27
PQS	15.23	1.79	42.40	42.71	42.87	54.19
PRS	7.25	4.43	16.81	16.16	18.23	27.22
			Medium firms			
PES	14.55	2.37	45.32	41.94	37.09	60.05
PPI	12.64	0.00	9.62	17.13	18.71	27.55
PQS	15.50	1.83	48.25	43.04	39.08	67.26
PRS	6.54	7.09	12.37	28.56	25.34	41.80
			Small firms			
PES	12.04	1.14	40.71	24.25	27.73	48.02
PPI	13.92	0.00	5.17	15.29	6.81	5.99
PQS	15.04	1.02	44.68	30.26	31.74	51.58
PRS	8.42	0.82	19.49	6.82	15.42	27.31

Note: The tail dependence coefficients between proxies and benchmarks (expressed as a percentage) are estimated for daily and monthly data within the Gumbel copula. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

Let us illustrate two cases: one, for which there exists a dependency between low and high-frequency measures, and another, for which this dependency is much weaker. Fig. 3 illustrates the dependence of two "competitive" pairs in daily (left) and monthly (right) frequency. In the top panel the pair of PQS and CQS is presented (with dependency coefficient $\lambda = 46.61\%$ for daily or $\lambda = 56.92\%$ for monthly data), while in the bottom panel PPI and CSA are displayed ($\lambda = 0.0\%$ or $\lambda = 21.47\%$, respectively). When these two panels are compared, the first pair illustrates clearly the case of a higher dependency than the second pair, regardless of frequency.

Each graph represents the estimated dependency coefficient for a pair of a benchmark and a proxy. The top row displays the Percent Quoted Spread (PQS) and the Closing Quoted Spread (CQS), while the bottom row shows the Percent Price Impact (PPI) and the Effective Spread Estimator (CSA). The left panel displays dependencies in daily data, while the right one in monthly data. The estimations come from the Gumbel copula.

Fig. 4 shows the dynamics of the dependency coefficients for all pairs of a benchmark and a proxy within the time. The graphs clearly confirm previous conclusions: the best pairs are these that include CQS as a proxy, and PES or PQS as a benchmark.

Each graph represents dynamics of the monthly dependency coefficient for a pair of a benchmark and a proxy. The abbreviations are as follows: PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). In the columns HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

We also consider the dependencies estimated in differently specified copulas. Table 5 shows the tail dependence coefficients for the Gumbel, Joe and Hüsler-Reiss copulas. A comparison of the values leads to a conclusion that there are no substantial differences between these three specifications as they indicate the same winner in the best proxy competition. For each pair the highest value is observed for the Hüsler-Reiss copula, while the lowest for the Joe copula. Regardless of copula taken into account, the winner in the best proxy competition remains the same.[4]

Finally, we conduct an asymptotic nonparametric independence test for the 90th and 95th percentile. The results are shown in Table 6. The null hypothesis stating that there is no extreme dependency between the proxy and the benchmark is rejected in five (90th percentile) and three (95th percentile) out of twelve cases. Thus, according to the test for 90th percentile the dependency in tails is found for the following pairs: HLR-PES, HLR-PQS, HLR-PRS, CQS-PES, and CQS-PQS. This result confirms earlier findings that CQS is the best proxy. Additionally, it also shows which of the other two is better: for HLR we find the dependence in extreme values

[4] Additionally we obtained surface densities of copulas for the winning pair of the benchmark and the proxy, PQS and CQS. The following estimates were obtained: in the Gumbel copula $\theta = 1.620$, in the Joe copula $\theta = 1.814$, and in the Hüsler-Reiss copula $\theta = 1.211$. As they look very similar, for the sake of brevity we show surface density of the Gumbel copula only in the Appendix (see Figure A3).

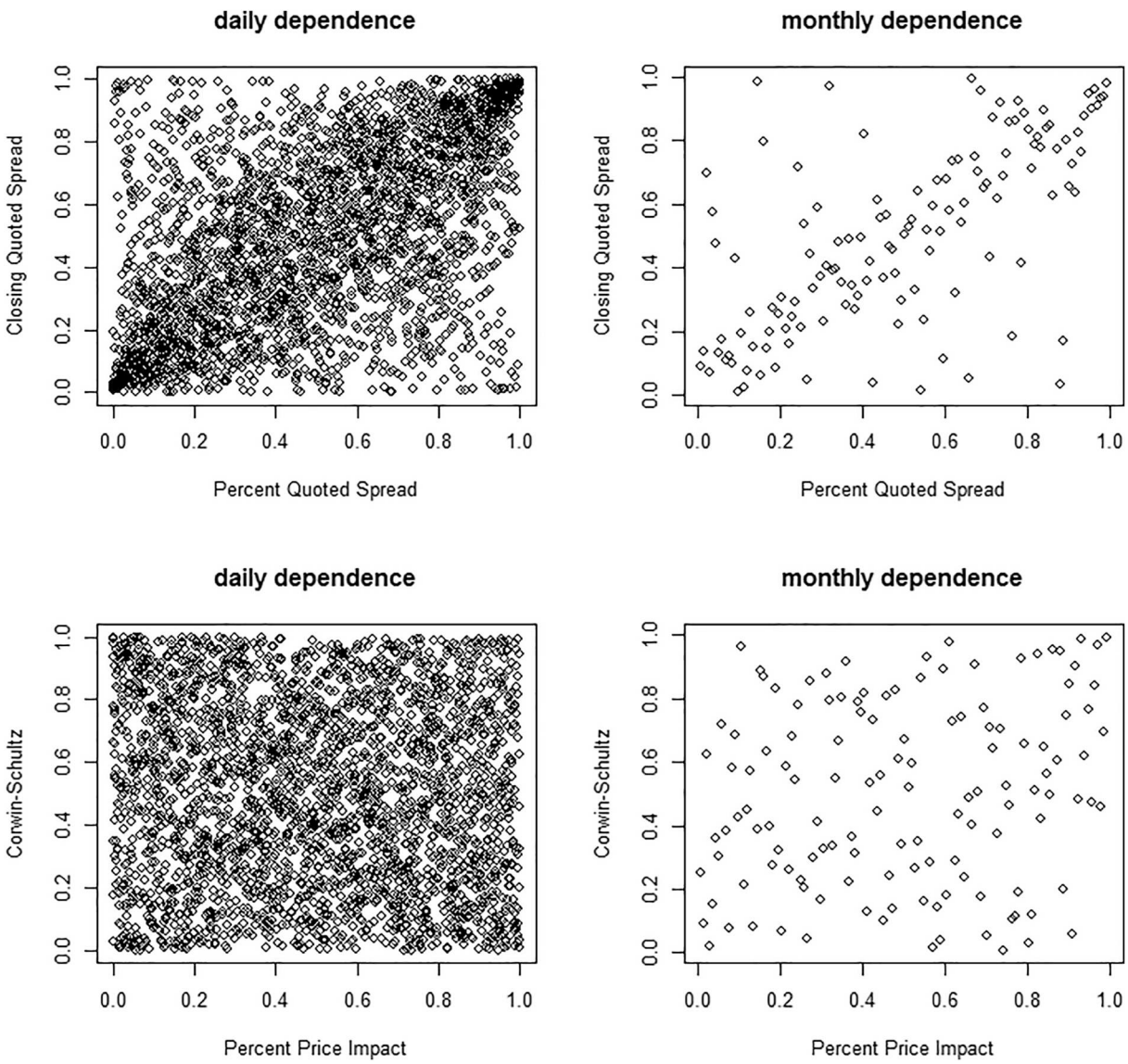

Fig. 3. Scatter plots of copulas for two pairs of liquidity measures.

with two benchmarks, while for CSA the test indicates no tail dependence with any benchmark. Thus HLR seems to be better than CSA.

4.3. The tail dependence in subperiods

We also examine the robustness of obtained results by calculating tail dependence coefficients in three different subperiods separately. Thus we check if changing the sample period could assign different winners. This is justified as there is a shift in benchmarks that is not accompanied by changes in proxies at the time of the UTP system introduction in 2013.

We obtain three almost equally long subperiods that allowed us to estimate the tail dependency coefficients. The first period is from 3.01.2006 to 21.08.2009 (908 days), the second is from 24.08.2009 to 12.04.2013 (914 days) and the third which lasting 914 days starts on the day when the UTP system was launched and ends at the end of 2016. Table 7 shows the tail dependence coefficients estimated for daily data in two approaches, i.e. cross-section and time-series. The coefficients in the subperiods are different, but the results are stable.

Our earlier findings are confirmed: the strongest dependency is observed for the CQS proxy with two benchmarks, that is PES and PQS, and that result is observed in both approaches. In the average cross-section the tail dependence coefficients are definitely higher than in the time-series approach. Although in the average cross-section both HLR and CSA proxies perform similarly, in the time-series HLR outperforms CSA substantially. These conclusions are in line with the previously obtained results for the whole sample.

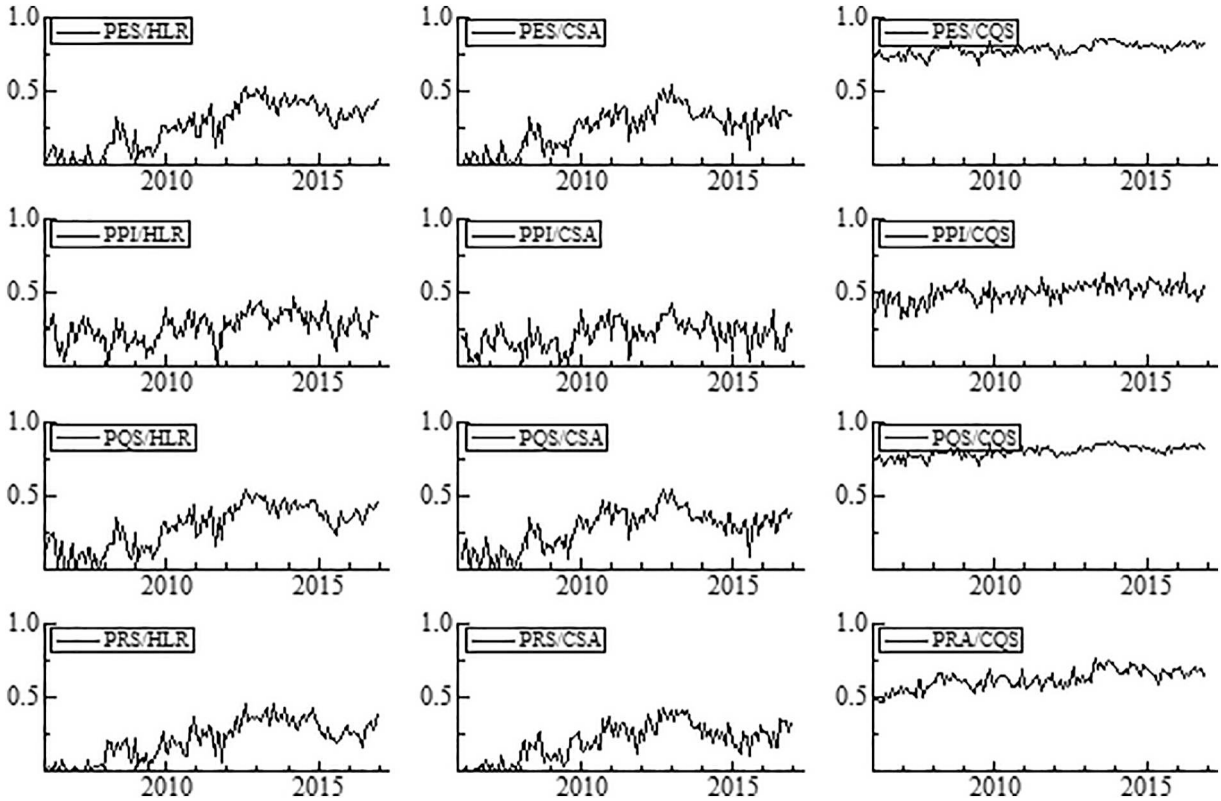

Fig. 4. Tail dependence coefficients for monthly data.

Table 5
Portfolio time-series: tail dependence coefficients estimated with three different copulas.

	Daily			Monthly		
	HLR	CSA	CQS	HLR	CSA	CQS
			Gumbel			
PES	16.83	0.22	43.52	39.81	44.23	55.62
PPI	16.11	0.00	6.25	21.91	21.47	16.26
PQS	18.12	0.21	46.61	43.01	46.60	56.92
PRS	8.59	1.46	19.84	21.03	27.54	35.68
			Joe			
PES	15.19	0.00	38.13	37.28	43.08	47.93
PPI	13.78	0.00	3.63	22.69	22.59	12.49
PQS	16.52	0.00	40.88	41.20	46.38	49.50
PRS	7.27	0.00	16.36	16.32	23.95	28.69
			Hüsler-Reiss			
PES	21.28	0.24	50.06	49.60	54.04	63.45
PPI	18.54	0.00	8.72	29.65	30.06	23.15
PQS	22.80	0.00	53.46	52.82	56.58	64.63
PRS	11.48	1.47	24.14	26.41	33.56	43.42

Note: The tail dependence coefficients between proxies and benchmarks are estimated for daily and monthly data via the Gumbel, Joe and Hüsler-Reiss copulas. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014). The tail dependece coefficients are expressed as a percentage.

5. Conclusions and further research

Our paper adds to a growing body of literature on liquidity benchmarks and proxies. Surveys such as that conducted by Karolyi et al. (2012) showed that when markets become turbulent the statistical and dynamic properties of liquidity measures change and the

Table 6
Asymptotic Neyman-Pearson independence test for time-series approach.

	HLR	CSA	CQS
		90th percentile	
PES	4.39 (0.00)	−2.26 (0.37)	6.03 (0.00)
PPI	−7.32 (0.61)	−1.49 (0.24)	−3.49 (0.26)
PQS	3.38 (0.00)	−0.30 (0.15)	6.26 (0.00)
PRS	−1.15 (0.04)	−3.23 (0.53)	−6.28 (0.20)
		95th percentile	
PES	1.08 (0.03)	−1.30 (0.54)	−3.25 (0.15)
PPI	−1.87 (0.61)	−1.34 (0.55)	−0.16 (0.17)
PQS	0.51 (0.05)	−1.81 (0.70)	−0.89 (0.03)
PRS	−0.75 (0.18)	−1.29 (0.58)	−7.60 (0.99)

Note: This asymptotic Neyman-Pearson independence test is conducted on daily data for 12 pairs that include one benchmark and one proxy. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014). The results for the 90th percentile are presented in the top panel (the statistic and *p*-value), while the results for the 95th percentile are given in the bottom panel.

Table 7
Tail dependence coefficients in three subperiods.

	Average cross-section			Time-series		
	HLR	CSA	CQS	HLR	CSA	CQS
			Subperiod: 3.01.2006–21.08.2009			
PES	13.36	14.45	54.37	15.71	0.00	29.56
PPI	20.70	9.65	30.18	13.79	0.30	5.77
PQS	16.61	16.47	56.64	20.00	0.00	31.69
PRS	11.93	14.04	43.89	12.86	0.00	13.45
			Subperiod: 24.08.2009–12.04.2013			
PES	26.77	27.64	64.30	14.09	0.85	38.23
PPI	25.79	14.05	34.05	22.56	0.30	11.75
PQS	29.06	29.73	66.78	13.04	0.73	41.92
PRS	24.14	26.82	54.62	5.08	2.72	13.07
			Subperiod: 15.04.2013–30.12.2016			
PES	27.78	22.16	68.67	17.71	3.54	48.41
PPI	26.08	14.10	40.63	9.26	0.00	0.00
PQS	30.28	23.79	72.52	20.06	3.54	52.49
PRS	25.77	22.10	59.86	11.00	5.18	27.54

Note: This table presents the tail dependence coefficients (expressed as a percentage) between benchmarks and proxies calculated for daily data in three subperiods, consisting of 908 (1st subperiod), 914 (2nd subperiod) and 914 (3rd subperiod) days. Four benchmarks and three proxies are considered. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

bid-ask spreads tend to rise rapidly. The importance of our study lies in the analysis of various percent-cost liquidity proxies and benchmarks when liquidity evaporates and transaction costs increase. The rationale for the analysis of extreme illiquidity comes from the fact that investors are more concerned about potential illiquidity when the market becomes more turbulent.

We examine how adequate liquidity percent-cost proxies, well known from the literature, prove to be when the market experiences extreme illiquidity. In the paper we focused on the dependency between proxies calculated from daily data and benchmarks obtained from high-frequency data. In order to focus on the extreme values, we applied the copula methodology. Both in the cross-section and the time-series approach the highest tail dependence was found for the Closing Percent Quoted Spread (CQS) of Chung and Zhang (2014). We contribute to the literature by finding that this proxy performs well also in the emerging market. It turns out to be the best proxy among percent-cost measures when liquidity evaporates. The use of such a measure saves a lot of computational burden inevitable when the high-frequency data are applied. Adding bid and ask quotes at the end of the day, required to calculate the CQS proxy, to the widely available daily databases would help obtain a much better approximation of the spreads and thus increase applicability of the datasets, as well as improve the accuracy of the liquidity proxies.

We also found substantial differences in the spreads between big and small companies; not surprisingly, for the former the spreads are lower than for the latter. However, the size of the company does not influence tail dependence coefficients for benchmarks and proxies. It indicates that the choice of the winning proxy holds across the stocks issued by companies from different size groups.

Empirical findings in this paper have potentially important implications for investors and other financial market participants. As the emerging markets suffer from illiquidity more often than the developed ones, it seems that the choice of the proxy ought to be particularly relevant. Investors are equipped with a proper measure of liquidity, which will allow them to control the impact of liquidity risk for stocks also during market crisis. Thus investment in the emerging markets becomes less costly and more attractive. The results are particularly useful for those, who need a simple and well-performing proxy for unobserved liquidity.

Acknowledgments

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Appendix A

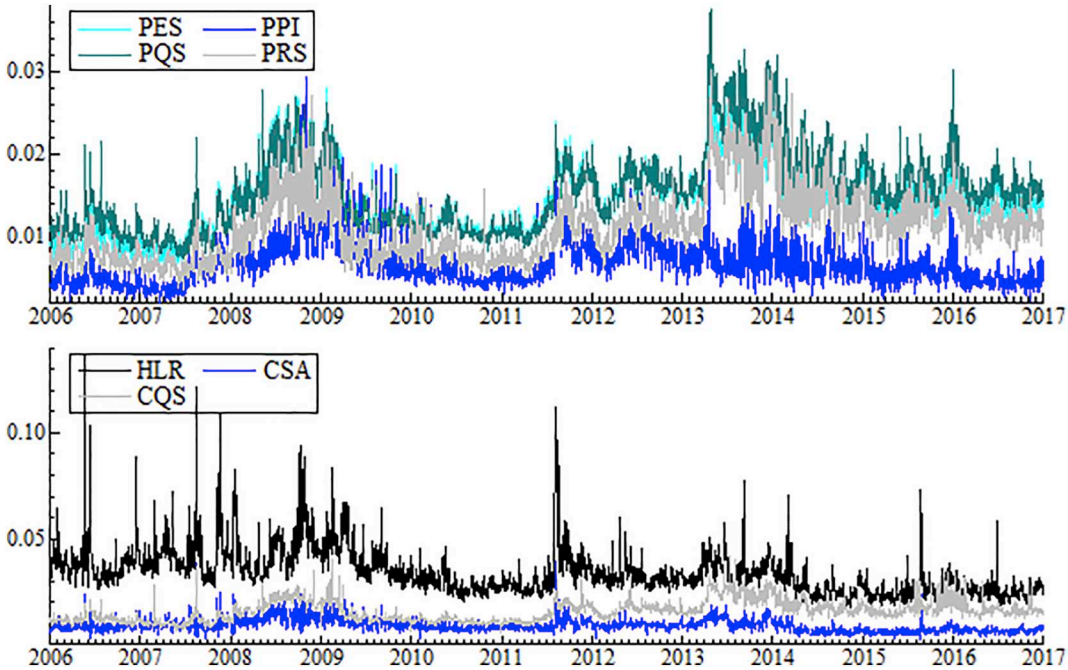

Fig. A.1. Daily spread measures.
The top panel shows four benchmarks obtained for a sample of stocks in daily frequency: PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). In the bottom panel HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014). This Figure shows both classes of liquidity measures in daily frequency: the top panel is for benchmarks, while the bottom is for proxies. Higher frequency allows us to observe some discontinuities in the measures: in particular there are the days, in which sharp jumps in liquidity appear, but they expire quickly.

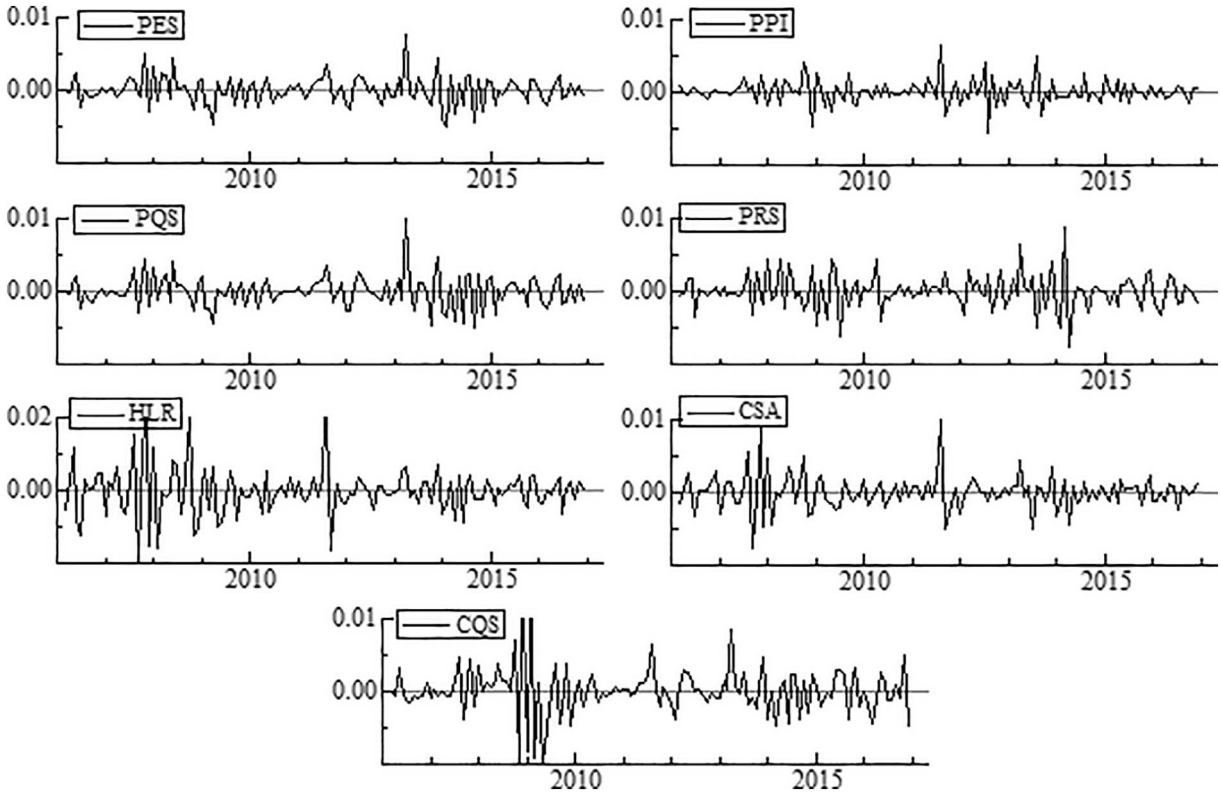

Fig. A.2. First differences of monthly liquidity measures.
The figure shows differences of four benchmarks obtained for a sample of stocks in monthly frequency: PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). The next three graphs are showing the first differences of the proxies: HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

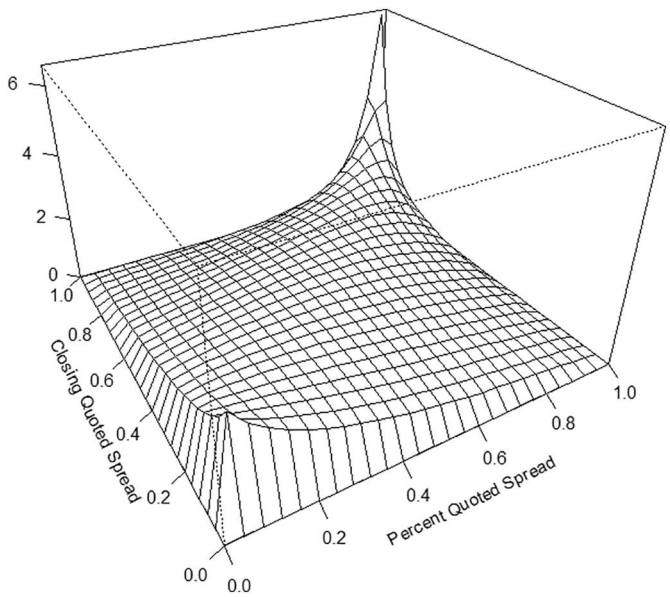

Fig. A.3. Surface density of the Gumbel copula for the Percent Quoted Spread (PQS) and the Closing Quoted Spread (CQS).

The figure presents surface densities of copulas for the Percent Quoted Spread (PQS) and the Closing Quoted Spread (CQS). The plot shows the Gumbel copula with estimated theta 1.620.

Table A.1
Correlation coefficients: Average cross-section.

	Daily			Monthly		
	HLR	CSA	CQS	HLR	CSA	CQS
			All stocks			
PES	24.59	17.88	71.18	27.70	23.60	88.65
PPI	26.98	9.74	40.40	31.68	24.51	60.79
PQS	27.68	20.11	73.71	31.62	27.38	89.88
PRS	19.12	16.03	59.44	18.58	16.44	72.50
			Big stocks			
PES	3.66	−1.35	66.24	−9.28	−14.53	90.59
PPI	16.06	1.65	40.39	4.02	−3.13	64.12
PQS	7.16	1.55	68.23	−5.25	−10.24	91.62
PRS	1.80	−0.16	52.46	−8.08	−11.80	75.79
			Medium Stocks			
PES	18.15	14.28	60.85	7.20	9.70	80.80
PPI	23.20	6.60	26.32	18.08	15.13	40.77
PQS	21.20	16.38	63.63	10.31	12.63	82.45
PRS	12.16	11.33	49.53	1.55	5.03	63.45
			Small Stocks			
PES	24.50	19.33	67.12	23.95	21.60	84.78
PPI	25.05	7.23	28.91	25.84	18.34	46.38
PQS	28.43	22.08	71.12	29.49	27.12	86.94
PRS	18.90	17.37	55.69	16.39	16.41	67.91

Note: The table presents correlation coefficients (expressed as a percentage) for liquidity benchmarks and proxies in an average cross-section. The coefficients are calculated both for daily and monthly data. Additionally three size groups are considered: big, medium and small. Among benchmarks PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhan (2014).

Table A.2
Correlation coefficients: Portfolio time series.

	Daily			Monthly		
	HLR	CSA	CQS	HLR	CSA	CQS
			All stocks			
PES	20.06	−0.36	43.51	42.23	53.16	48.55
PPI	14.19	−7.10	2.66	36.78	35.44	7.31
PQS	22.24	0.81	45.77	45.40	55.53	51.25
PRS	9.35	−1.12	16.76	13.27	27.31	28.91
			Big stocks			
PES	19.71	0.72	24.61	53.78	52.15	52.29
PPI	11.87	−1.84	4.71	24.05	21.13	19.53
PQS	20.95	−0.29	27.13	51.07	51.51	52.69
PRS	11.23	2.77	10.13	18.11	18.57	27.07
			Medium Stocks			
PES	14.44	4.99	61.91	48.73	44.15	73.14
PPI	12.51	−4.19	3.74	17.63	16.58	27.72
PQS	16.08	5.12	6.34	48.72	44.82	80.94
PRS	8.19	8.59	9.54	28.32	27.40	32.12
			Small Stocks			
PES	13.72	0.54	35.54	23.93	37.69	24.54
PPI	10.19	−3.61	−0.41	26.29	11.17	−10.69
PQS	17.02	1.60	37.82	30.29	41.55	29.22
PRS	9.10	−2.09	16.97	−1.82	23.17	13.28

Note: The table presents correlation coefficients (expressed as a percentage) for liquidity benchmarks and proxies in the time-series approach. The coefficients are calculated for daily and monthly data. Four benchmarks are considered: PES stands for the Percent Effective Spread, PPI is the Percent Price Impact, PQS denotes the Percent Quoted Spread, while PRS is the Percent Realised Spread (Fong et al., 2017). Among proxies HLR denotes the High-Low Range (Będowska-Sójka, 2019), CSA stands for the Effective Spread Estimator of Corwin and Schultz (2012), while CQS is the Closing Percent Quoted Spread of Chung and Zhang (2014).

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Reading Comprehension Questions

1. What two types of measures do the authors mention for observing liquidity in a market?

   The authors mention **benchmarks**, which are calculated using high-frequency data, and **proxies**, which are calculated using low-frequency data (page 1).

2. Why are some investors limited to using proxies for observing liquidity?

   Some investors **do not have access to the high-frequency data** needed to calculate a benchmark, so they must use a proxy (page 1).

3. What market did the authors focus on in their analysis?

   The authors used data from the **Warsaw Stock Exchange** between January 2006 and December 2016 (page 4).

4. Which proxy did the authors find performed best in low-liquidity environments?

   The authors found that the **Closing Percent Quoted Spread (CQS)** was "the best proxy among percent-cost measures when liquidity evaporates" (page 13).